FAKE NEWS DETECTION USING NLP AND LINGUISTIC FEATURE ANALYSIS

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INTRODUCTION

- Online news and social media have become major sources of information.
- The rapid spread of misinformation makes it difficult to distinguish genuine news from fabricated content.
- Manually verifying large volumes of news articles requires considerable time and effort.
- Natural Language Processing (NLP) enables computers to analyse and process human language.
- Machine learning can identify patterns in news articles and classify them using previously labelled examples.

Project Aim: To develop a machine learning approach for classifying news articles as fake or real using textual and linguistic information.

PROBLEM STATEMENT

- Traditional text classification approaches analyse words and phrases in news articles. However, linguistic characteristics such as sentence length, punctuation, vocabulary diversity, and sentiment may also provide useful information.
- This project investigates whether combining these linguistic characteristics with TF-IDF representations improves classification performance.

OBJECTIVES

Research Approach

- Collect and preprocess fake and real news articles.
- Perform exploratory data analysis on the dataset.
- Extract textual representations using TF-IDF.
- Calculate linguistic features from news articles.
- Train and compare different machine learning models.
- Evaluate performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- Research Question: Does combining linguistic features with TF-IDF improve fake news classification compared with TF-IDF alone?

LITERATURE REVIEW

- **Study 1: Linguistic Features-Based Framework for Automatic Fake News Detection (2022)**

  Research focus:

  - Investigated linguistic features for automatic fake news detection.
  - Examined 26 linguistic features, including readability, complexity, and writing-style characteristics.
  - Compared TF-IDF, Count Vectorizer, and Hash Vectorizer.
  - Evaluated multiple machine learning models across four datasets.

  Key finding:
  The study demonstrated that combining linguistic feature analysis with machine learning can support fake news classification. Performance varied across datasets.
  Relevance to Our Project:
  This research supports investigating linguistic features alongside TF-IDF representations.
  Reference: Computers & Industrial Engineering, 172, 108432 (2022).

LITERATURE REVIEW

- **Study 2: An Empiric Validation of Linguistic Features in Machine Learning Models for Fake News Detection (2023)**

Research focus:

- Investigated 88 linguistic features.
- Grouped features into surface information, part-of-speech, discourse, and readability categories.
- Evaluated linguistic features using a dataset containing 42,677 news articles.
- Identified 32 features that showed significant differences between fake and real news in the studied dataset.

OUR APPROACH

Compare:

- TF-IDF with Logistic Regression.
- TF-IDF with Linear SVC.
- Linguistic features alone.
- Combined TF-IDF and linguistic features.
- Reference: Puraivan, Venegas, and Riquelme, Data & Knowledge Engineering, 147 (2023), Article 102207.

Methodology

- **DATASET AND PREPROCESSING**

ISOT Fake and Real News Dataset
Two CSV files are used:

- Fake.csv — articles labelled as fake.
- True.csv — articles labelled as real.

- **Preprocessing Steps**

1. Load both CSV files using Pandas.
2. Assign labels: Fake = 1 and Real = 0.
3. Combine both datasets.
4. Check missing values and duplicate articles.
5. Remove duplicate records where appropriate.
6. Examine class distribution and article lengths.
7. Split the data into training and testing sets using an 80:20 ratio.

Important: Fit the vectorizer and feature scaler on the training data only to prevent test-data leakage.

Methodology

- **FEATURE EXTRACTION**

  1. TF-IDF Representation

  Term Frequency–Inverse Document Frequency (TF-IDF) converts text into numerical vectors.

  - Unigrams: Individual words.
  - Bigrams: Sequences of two consecutive words.
  - Assigns importance to terms based on their occurrence in a document and across the collection.

- **Feature Combination**

  1. Combined Representation = TF-IDF Features + Scaled Linguistic Features
  2. The aim is to determine whether these additional features provide useful information beyond TF-IDF alone.

Methodology

- **MODEL TRAINING AND EVALUATION**

  - Accuracy: Overall proportion of correct predictions.
  - Precision: Proportion of predicted fake articles that are labelled fake.
  - Recall: Proportion of labelled fake articles correctly identified.
  - F1-score: Harmonic mean of precision and recall.
  - Macro-F1: Average F1-score across both classes, giving each class equal weight.
  - Confusion Matrix: Displays correct and incorrect predictions for both classes.

Experimental Goal: Compare all four configurations using the same held-out test set.

Conclusion

- Developed a machine learning workflow for fake news classification.
- Used TF-IDF to represent article text numerically.
- Extracted linguistic features such as word count, sentence length, punctuation usage, lexical diversity, and sentiment polarity.
- Compared text-based models, linguistic-feature-based classification, and a combined approach.
- Used standard evaluation metrics to assess model performance.

Thank You